

Anurag Pradhan

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

SUMMARY

Prefinal-year Computer Science student at VIT Chennai with hands-on experience in Deep Learning and Machine Learning. Skilled in building and deploying efficient CNN, Transformer, and Vision-Language models, along with PyTorch experimentation, data preprocessing, optimization, and on-device inference. Published researcher (IJCNLP-AACL 2025) focusing on accessibility-centered evaluation of lightweight VLMs. Strong data science foundation, including feature engineering, exploratory analysis, and supervised learning. Adept at delivering practical, production-ready ML solutions within engineering teams.

EDUCATION

- Vellore Institute of Technology** Chennai, India
Bachelor of Technology in Computer Science; GPA: 7.74/10 Aug 2023 – June 2027
- Sri Venkateswara Jr. College** Visakhapatnam, India
Class 12th (MPC); Score: 88.4/100 July 2021 – June 2023
- Sri Chaitanya Techno School** Visakhapatnam, India
Class 10th Score: 93.6/100 Feb 2020 – Mar 2021

SKILLS SUMMARY

- Programming Languages:** Python, C, C++, Java
- Core Expertise:** Deep Learning, Computer Vision, CNNs, Vision Transformers, Image Processing, Sequence Models (LSTM/GRU/RNN), Generative AI, Large Language Models
- Frameworks & Libraries:** PyTorch, Hugging Face Transformers, LLaMA, BEiT, BLIP, Llama.cpp
- Tools & Platforms:** Git, Docker, CUDA, ffmpeg, Linux
- Soft Skills:** Leadership, Technical Writing, Event Management, Time Management

EXPERIENCE

- National Aluminium Company Ltd (NALCO)** Damanjodi, Odisha
Computer Science Intern June 2025 – July 2025
 - Contribution:** Worked on RespiGuard AI, an AI-based respiratory safety system for mining environments. Improved hazard detection reliability using deep-learning-based visual analysis.
 - Tech Stack:** Custom YOLOv5, PyTorch, Data Augmentation, Model Evaluation
- Team Ignition, VIT Chennai** On Site
Flight Software Lead Apr 2024 – Present
 - Contribution:** Developed flight software for model rockets, including fin actuator control, stability algorithms, onboard video stabilization, and Kalman-based state estimation.
 - Tech Stack:** Arduino, C/C++, PID Control, Image Processing, Deep Learning

PROJECTS

- Liver Disease Prediction:** Developed a YOLOv5-based detection framework with Stable Loss, augmentation strategies, and Adam optimization. Added interpretability through heatmap visualizations. [\[GitHub Repo\]](#)
Tech: Python, PyTorch, YOLOv5, CUDA
- Brain Tumor Segmentation:** Built a BEiT-transformer segmentation pipeline with skip connections and attention-driven decoding. Trained with Albumentations and CE-Dice hybrid loss for stable convergence. [\[GitHub Repo\]](#)
Tech: Python, PyTorch, Vision Transformers, Semantic Segmentation
- JanaSathi: Odia E-Governance Chatbot:** Designed a bilingual RAG-based chatbot for Odisha government schemes using OpenRouter LLMs and MiniLM embeddings. Integrated Odia translation with Cohere Command-R+. Achieved 2nd place in AMD-sponsored OdiaGenAI Hackathon 2025. [\[GitHub Repo\]](#)
Tech: Python, Transformers, LangChain, Retrieval-Augmented Generation
- Mobile Deployment of Small Vision-Language Models:** Deployed SmolVLM2 (2.2B & 500M) Video-Instruct models on-device (Vivo Y27) using llama.cpp with FP16 and Int8 optimization. Demonstrated real-time video captioning within strict mobile constraints. [\[GitHub Repo\]](#)
Tech: VLMs, Quantization, Llama.cpp, ffmpeg, Termux, Kotlin

CONFERENCE PUBLICATIONS

- Towards Blind and Low-Vision Accessibility of Lightweight VLMs and Custom LLM-Evals:** S. S. Baghel, Y. P. S. Rathore, A. Pradhan. Accepted at the 14th International Joint Conference on Natural Language Processing & the 4th Asian Chapter of ACL (IJCNLP-AACL 2025), Mumbai, India.